



Form Version: 19 December 2023

A1. Application details

Title of the project: SMASH: Synthesis and Multimodal Analytics System for Humanities
Duration of the project (in months): 12 months

A2. Public summary

English public summary:

Our project, SMASH, is working on a new way to understand how people communicate with each other beyond words. We are focusing on how people use their body language, facial expressions, and tone of voice to express themselves when they talk. We want to extend an already-developed tool that tracks body movement and masks audio and video for privacy. The extension would be to automatically capture and analyze non-verbal and verbal communication aspects to augment researchers and clinical professionals studying communication. This would be a timely achievement in social science supported by computer science.

Words: 94/100

Dutch public summary:

Ons project, SMASH, werkt aan een nieuwe manier om te begrijpen hoe mensen met elkaar communiceren buiten woorden om. We richten ons op hoe mensen hun lichaamstaal, gezichtsuitdrukkingen en stem gebruiken om zich uit te drukken. We willen een reeds ontwikkelde tool uitbreiden die lichaamsbewegingen volgt en audio en video maskeert voor privacy. De uitbreiding zou bestaan uit het automatisch vastleggen en analyseren van non-verbale en verbale communicatieaspecten om onderzoekers en klinische professionals te helpen bij het bestuderen van communicatie. Dit zou een tijdige innovatie zijn in de sociale wetenschap ondersteund door computerwetenschap.

Words: 94/100

A3. Budget

Type of costs	Clarification	Costs in euros
Personnel	Third party hire computer science institute (postdoc level)	45.000
Travel and accommodation	Attendance at conferences and workshops	5.000
Own contribution (optional)		N/A
Total request from NWO		50.000

A4. Budget Justification

We allocate €45.000 for a third-party hire on a post-doctoral level at a European computer science (CS) research institute. The external postdoc is crucial for the success of this project as this person has spearheaded the development of the software base. The postdoc level is required as there is a need for broad conceptual and technical knowledge spanning social and computer science. The postdoc will spearhead the core day-to-day developments of SMASH under the supervision of two established Dutch and European researchers over a year (part of the position is already funded; this grant would support 1.0fte for seven months). Travel and accommodation costs, amounting to €5.000, are split over conferences and workshops (€1.500) and €3.500 for travel and accommodation between the Netherlands and the European city.

Words: 149/150

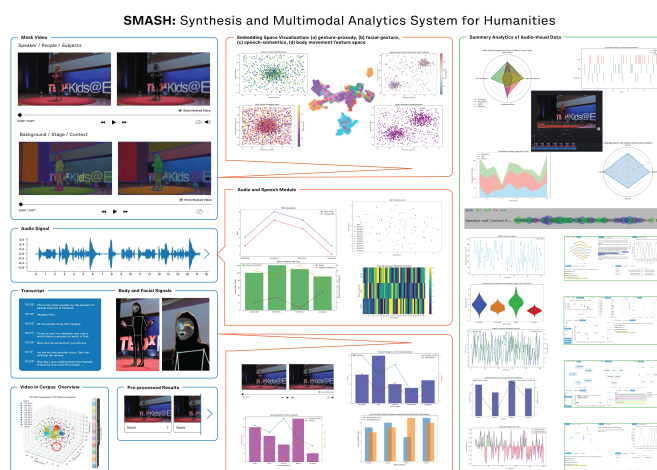
Section B. Scientific proposal

B1. Scientific proposal

Bodily gestures and tone of voice underline that the full range of human expression makes use of more than words alone [1]. How to study this extra layer of communication in an integrated way is a challenging problem in social science [1,2], requiring technical innovation guided by social science. That's where the SMASH project comes in. SMASH will be a fully open-source tool that helps us better understand the composition of non-verbal communication aspects, like body language, facial expressions, and the melodic aspects of speech. It will combine an already developed masking tool that masks audio-visual data to protect people's privacy with analysis modules that inform about how speakers communicated beyond the words uttered, thereby serving as a general tool for data-archiving and behavioral analysis of audiovisual data in social sciences in humanities. SMASH aims to better integrate the understanding of a rich set of communicative cues by bridging the gap between computer science and social science with far-reaching applications. For example, clinicians can eventually use SMASH to quantitatively monitor more holistic aspects of communication that deteriorate in communicative disorders like Aphasia [3]. Seeing the rapid adoption of CHATGPT, gaining insights into the complete spectrum of human emotions, actions, and non-verbal communication is of utmost importance for comprehending *language in the wild* and developing genuinely relatable and socially adept chatbots and artificial agents.

Social science research has more opportunity to benefit from computer science advances that enable the simultaneous processing of multiple communicative signals [4–6]. For example, the field of computer vision now enables automatic facial and body movement tracking from video, which can then form input for automatic classification of behaviors like pointing with the hand, smiling, etc. [7]. Our team is ready to implement such computational pipelines [1,8] into an integrated toolkit, as Figure 1. explains.

Figure 1. Example SMASH results dashboard. SMASH would take as input audio-visual recordings of (a) communicating individual(s) and a summary of their multimodal communication characteristics as output. For instance, it shows the number and diversity of hand gestures [2], changes in posture and facial expressions relevant to communication context [8], and how voice tone changes with other gestures [9]. It would also capture several trouble moments, such as pauses and fillers in speech [10] and summarize what is being discussed using automatic speech transcription and semantic analyses [10]. By integrating these dimensions - body language, facial expressions, voice, and content - in an easily understandable way, SMASH allows a detailed zoomed-out view of the full gamut of communication modes. SMASH will also offer standard baselines based on already analyzed speaker databases (see goal 2 below). In this way, users of SMASH can conclude that a speaker “gestured more relative to a database of Dutch speakers” or “used more variation in their tone of voice, even when compared to a database of highly appreciated speakers”. The tool would also offer options for data export for customized analysis. We aim to make SMASH a versatile and easy-to-use tool for users without programming experience. Through initial testing and adaptation by social scientists, we aim for SMASH to become an indispensable tool in research and clinical contexts.



This project's first goal is to develop an accessible communication analysis toolkit called SMASH, as outlined above. This toolkit will be integrated with software that performs masking and body motion tracking (already developed by the current team). The second goal is to unleash SMASH to address the following research question: how do these multiple communicative signals structurally interrelate, if at all, in audiovisual recordings in speakers from diverse backgrounds?

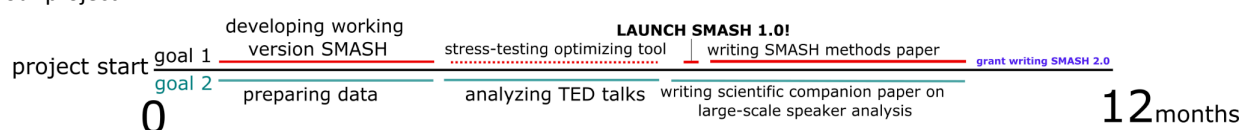
GOAL 1 – SMART formulation: Specific: Since we already have a codebase that processes audiovisual data with a user interface, we will integrate new modules within that codebase. Modules will include those that analyze melodic features of speech (prosody) [12] and assess semantic content (using so-called text-embedding pipelines, e.g., BERT [10]). For manual gesture analysis, we leverage GestureMap [4], which analyzes all the hand gesture movements to look for recurrent gestures. Finally, we will implement facial expression clustering using EmoCo [11]. After integrating these modules, SMASH will further implement data science visualization to maximize the value and summaries of the high-dimensional data [12]. Measurable: We will launch new beta versions every time we have integrated a new analysis module; as such, our progress can be incremental with value-added along the way. Attainable: The modules we want to add are based on open-source software already developed and tested but never integrated for multimodal analytics into a single tool. We can work stepwise, first implementing the most valuable modules (e.g., facial and hand gestures). Timely: After 3 months, we will have at least three behavioral analysis modules integrated into a complete beta version of SMASH that we can use for scientific analysis (goal 2). In 6 months, the official release version will follow, where we will have a fully functional and stress-tested masking, tracking and communication analysis toolkit that can be widely released to external users.

GOAL 2 – SMART formulation: Specific: Drawing on SMASH, we will analyze the growing corpus of public speaking audio-visual recordings of TED Talks, comprising > 3,000 speakers across several languages, including Spanish, German, and Arabic. We will assess the multimodal linguistic diversity by analyzing this corpus using SMASH, which will also create a baseline for SMASH. Considering a diverse range of topics, speakers, and countries, we aim to see if common multimodal patterns stratify along languages, which would provide an understanding of the possibility of loosely-knit ‘multimodal’ languages. The sheer scale of our analyses is novel and could provide the first quantitative evidence that languages are spoken in a whole-body manner, including manual and facial gestures. Measurable: To test this, we will apply clustering goodness-of-fit metrics to assess whether certain combinations of multimodal summary statistics provided by SMASH fit categories of language spoken or geographical region. Attainable: The team has access to GPU-supported computing clusters that can help mine and preprocess the audiovisual data. The analyses can proceed in a modular fashion, such that first, English and Spanish languages are compared, followed by more languages; this way, research output can be generated incrementally. Timely: While we develop the first version of SMASH (3 months), we will simultaneously prepare the corpus from the start of the project by going through the motion tracking procedures already in place in our tool (motion tracking is used as input for communication analysis modules). As soon as we have a first working version, we can further analyze the data, which will take 5 months to generate the first results.

Feasibility

SMASH’s ethical benefits for a responsible use of AI. As interdisciplinary social and computer scientists we find it essential to mention that the data provided by SMASH is intended to be interpreted by a human in the context of specific research or applications. We thereby respect the nuances of communication that vary greatly depending on cultural, situational, and individual factors. SMASH does not make direct inferences about an individual’s emotional state based solely on communication patterns. Instead, it provides bottom-up data summaries on various aspects of communication, leaving the interpretation to users. This approach empowers humanities researchers to apply their expertise and contextual knowledge when interpreting the multimodal communication characteristics instead of leaving it to AI. These core ethical principles guide our project, ensuring that SMASH becomes an empowering resource for augmenting the analysis of human communication.

Team composition and risk management: An interdisciplinary team of socio-behavioral researchers from Dutch universities and a European computer science research institute has made progress in co-developing the privacy masking and body tracking component of SMASH. Thus, the codebase is already implemented. The team has also developed open-access pipelines for semantic network analytics and natural language processing of written text. Thus, SMASH is feasible and does not require recruiting new team members. In SMASH, we further leverage the integration of existing open-source tools, which, when combined, provide a new ground-breaking communication analysis tool. The interdisciplinary SMASH team can progress in this project piecemeal, prioritizing low-hanging fruit with high analytic gains (e.g., gesture and speech analysis). The human-in-the-loop approach further prevents potential risks of overzealous misuse of AI-implemented software. Finally, the team has already shown to be effective in developing computer science methods in social science in productive collaboration with social science university partners, thereby reducing the associated project risks for deploying SMASH. Below is the timeline we envision for our project.



Section C. Reference list

C1. Reference list

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